

A Novel CC-II Based Current Comparator and Its Application as Current mode Flash ADC

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Abstract: In this paper, a new design for current comparator based on Current Conveyor-II (CC-II) is proposed. The proposed current comparator utilizes the concept of positive feedback. Simulations have been performed on Pspice using 0.18 μ m CMOS technology with 1.8V supply. Final results confirm a fairly quick response, less power dissipation and a resolution of 4.4 μ A for the current comparator. A 2-bit current mode flash ADC is designed using the proposed comparator.

Keywords: Current mode, Current conveyor-II (CC-II), Current Comparator, resolution, propagation delay, ADC

I. INTRODUCTION

Nowadays, current mode circuits are gaining wide-spread preference over voltage mode circuits due to their numerous advantages. They operate at high speed, require low supply voltage and consume less power. [1] There are a wide variety of current mode building blocks that find application in various signal processing systems, amongst which, current comparators are quite popular.

Current comparators find applications in current mode systems such as ADCs and signal processing circuits. Many current comparators have been proposed and implemented in the literature, which can be categorized as current mirror based [2-3], Traff based [4-5],[10] and active elements based [6-7].

First category of current comparators based on current mirrors finds applications in such areas where high speed operation is not required. It utilizes high output resistance current mirrors to amplify the difference of input current and this difference is represented in terms of output voltage. Due to this high output resistance, frequency response deteriorates when inverters are added at output to achieve full swing.

Next category is that of a class of current comparators whose concept was first conceived by Traff [4]. The current

comparator reported by Traff [4] has improved performance in terms of speed, power dissipation and chip area, as compared to current mirror based comparator design. It utilizes feedback concept which restricts input node from slewing rail to rail.

Third category of current comparators is based on active elements such as current conveyors, as reported in [6-7]. Use of active elements shortens the switching time of the comparator. Further, the comparator is capable of operating at higher frequencies.

In this paper, a new structure of current comparator belonging to the third category has been proposed. This current comparator uses CC II as its core.

The paper is organised as follows: Section 2 introduces basics of CC II. Section 3 describes circuit design of proposed current comparator. Section 4 presents application of comparator as current mode ADC. Section 5 presents simulation results. Finally, conclusions are drawn in section 6.

II. BASICS OF CC II

A current conveyor is an electronic amplifier with unity gain. It is a three terminal device. There are three generations of current conveyors viz. CCI, CCII, CCIII. Of all these, CC-II is the most versatile. General symbol of a CCII is shown in Fig. 1.

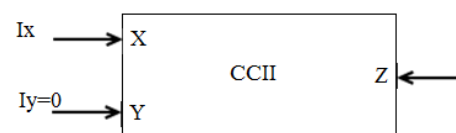


Fig. 1 General symbol of CCII

CCII is described by a matrix-

$$\begin{bmatrix} I_y \\ V_x \\ I_z \end{bmatrix} = \begin{bmatrix} 0 & 0 & 0 \\ \alpha & 0 & 0 \\ 0 & \pm\beta & 0 \end{bmatrix} \begin{bmatrix} V_y \\ I_x \\ V_z \end{bmatrix}$$

$$I_y=0, V_x=\alpha V_y, I_z=\pm\beta I_x \quad (1)$$

Terminal Y has infinite input impedance. Voltage at X follows that applied to Y. Thus X has zero input impedance. Current supplied to X is conveyed to Z which is high impedance output terminal. Thus current conveyor has the ability to act as a voltage buffer between its inputs and has the ability to convey current between two ports at extremely different impedance levels.

The CC-II can be considered as an ideal transistor [1] with-

$$Y = \text{Gate}, X = \text{Source}, Z = \text{Drain}$$

To verify this point, consider NMOS transistor shown in Fig. 2 below. Here the gate terminal would behave as an open circuit (as the conveyor terminal Y), the source terminal would exhibit zero input impedance (just as the conveyor terminal X). A current injected at the source would be conveyed to drain where the impedance level would be infinite (just as terminal Z in conveyor). It verifies that an ideal transistor behaves as a negative current conveyor. To create a more ideal transistor, in negative feedback of an op-amp, we place NMOS transistor as shown in Fig. 3. Thus giving a CCII- with reasonably good performance. But in this realization, current will flow only out of terminal X. Another realization can be obtained by placing a PMOS transistor in negative feedback loop of op-amp, where current will flow only into the terminal X.

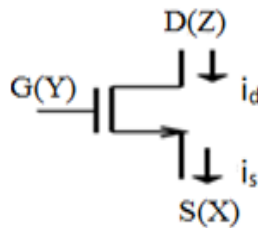


Fig 2 NMOS transistor

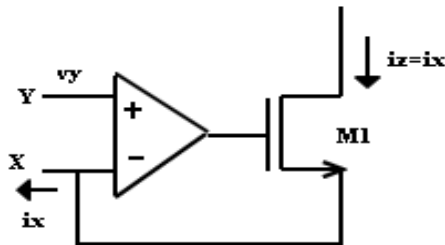


Fig. 3 Negative current conveyor using a 'super-transistor'

It means that a CCII realization can be obtained, by placing a complementary pair of MOS transistors in op-amp feedback path that allows bidirectional current flow as shown in Fig. 4. Transistors M1 and M2 act as buffer. Subsequently, this current is transferred to the output node via a complementary pair of current mirrors, where MP1 and MP2 are forming one pair of current mirror and MN1 and MN2 are forming other pair of current mirror.

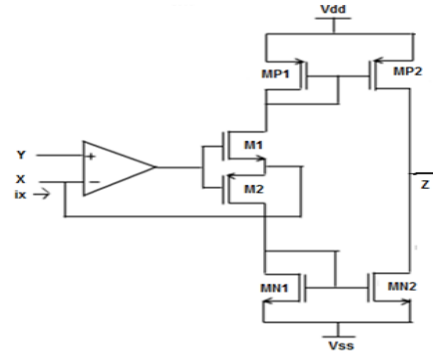


Fig. 4 Positive current conveyor

III. CIRCUIT DESIGN OF PROPOSED CURRENT COMPARATOR

A current comparator provides current to voltage conversion by comparing an input current to a reference current. Fig.5 shows the general block diagram of current comparator which consists of three stages. First stage is current subtractor stage which subtracts reference current (I_{ref}) from input current (I_{in}) and gives current difference. This current difference then goes to gain stage which gives amplified output in the form of voltage. In the gain stage, op-amp and current mirrors are used. Last stage is output stage which is a combination of inverters. This stage provides full voltage swing at the output [8].

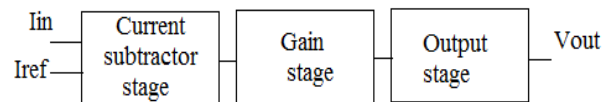


Fig.5 Block diagram of a current comparator

The current subtractor stage is shown in Fig.6. It accepts two input currents namely I_{in} and I_{ref} , and uses current mirrors to generate difference current $I_{in} - I_{ref}$. Various architectures for these subtractor circuits, to be used in conjunction with the current comparators, have been proposed [12]-[13]. We have used basic p-type current mirror based current subtractor in our implementation [11] and difference current is shown across resistance R.

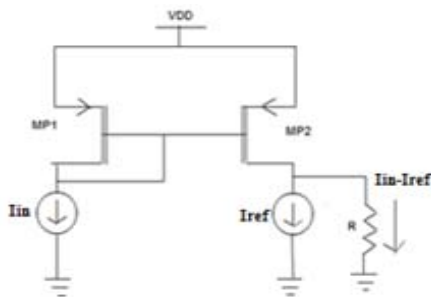


Fig. 6 Current subtractor stage

The second stage is the gain stage which uses CCII. CCII is used in stage, as expressed in [6], due to the low input impedance at node X and the inherent current to voltage conversion property of the CCII circuit. Fig. 7 shows the circuit of proposed current comparator.

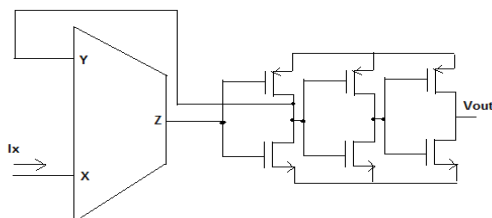


Fig.7 Proposed current comparator

Fig. 8 shows detailed structure of current conveyor comparator.

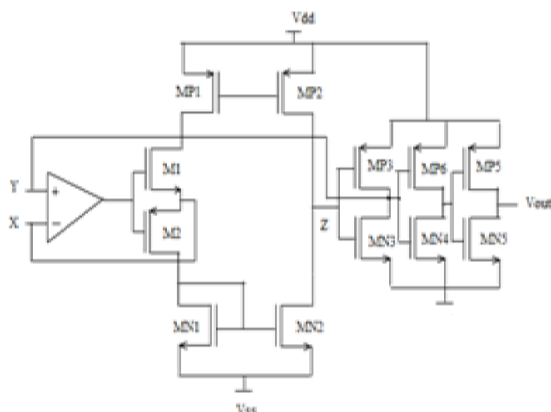


Fig.8 Detailed structure of current conveyor comparator

Fig. 9 shows internal diagram of CMOS Op-Amp as employed in our CC-II based current comparator.[9]

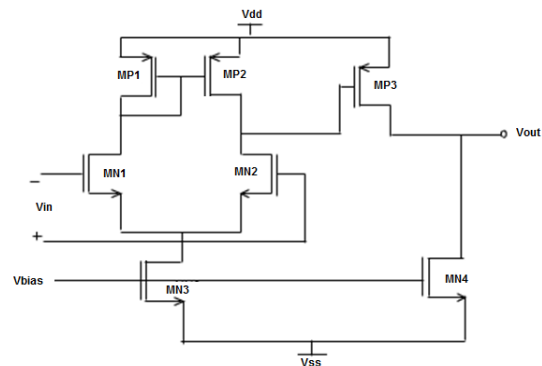


Fig.9 Internal diagram of CMOS Op-Amp

In this implementation, the difference current obtained from the current subtractor stage is injected into the terminal X which leads to a proportional voltage at terminal X. As per described principle of CCII, any change in input current leads to X-node variations and exact variation will appear at Z terminal. Opposite variation will appear at output of first inverter.

In order to decrease the response time of the circuit and to increase the sensitivity of input current variations, positive feedback network is used i.e. a positive feedback from the output of one of the inverters is fed to the terminal Y. Due to positive feedback this variation is conveyed at node Y which results in increase change in node X. Hence Vout will be obtained according to Iin.

The inverters in the output stage ensure full swing. Three inverters are used in this implementation, however any number of inverters can be used depending on whether full swing is obtained or not.

IV. APPLICATION OF PROPOSED COMPARATOR AS CURRENT MODE FLASH ADC

Comparator is an important part of ADC. So the above comparator is utilized in designing current mode ADC. The basic block diagram of 2-bit current mode flash ADC[14] is shown in Fig. 10. It consists of current subtractor, $2^{(n-1)}$ comparators i.e. 3 comparators as 2-bit and one 3×2 encoder. Fig. 6 is used to implement current subtractor and Fig. 8 is used to implement current comparators. 3×2 encoder is described in section 4.1.

ADC is designed to work in the current range of 0 – 60uA and reference currents are taken as 20uA, 40uA and 60uA for the three comparators respectively.

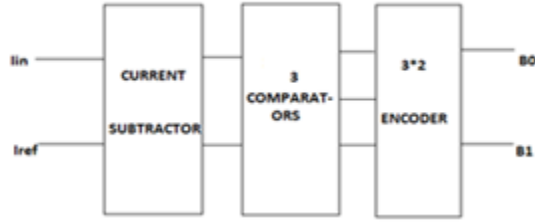


Fig.10 Block diagram of 2-bit current mode flash ADC

A. Encoder

3X2 encoder is required for 2 bit ADC shown in figure 2.2.3. It is used to convert three outputs of comparators to 2 bit digital output. According to the three current comparator outputs, the Encoder [15] indicates whether each one of $B1$ and $B0$ is 1 or 0.

TABLE I. ENCODER OUTPUT TABLE

Comp3	Comp2	Comp1	B1	B0
0	0	0	0	0
0	0	1	0	1
0	1	1	1	0
1	1	1	1	1

In the table I Comp1, Comp2 and Comp3 represents the corresponding output from the three comparators and B1 and B0 represents 2- bit digital output.

$$B1 = (\text{Comp1}) \text{AND} (\text{Comp2})$$

$$B0 = (\text{Comp1}) \text{AND} ((\text{Comp2}) \text{XNOR} (\text{Comp3}))$$

These expressions are used to implement 3*2 encoder design shown in Fig. 11

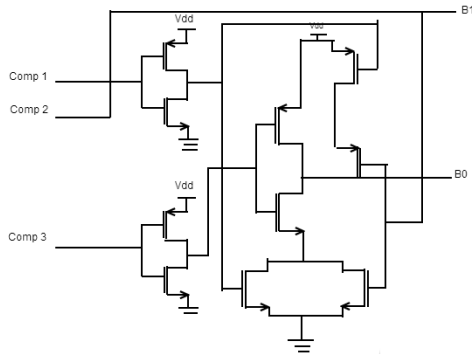


Fig.11 3*2 Encoder design

V. SIMULATION RESULTS

P-spice simulation results of proposed current comparator are presented using 0.18 μm technology using power supply of 1.8V. Proposed comparator shows a resolution of 4.4 μA . Number of transistors used are 19.

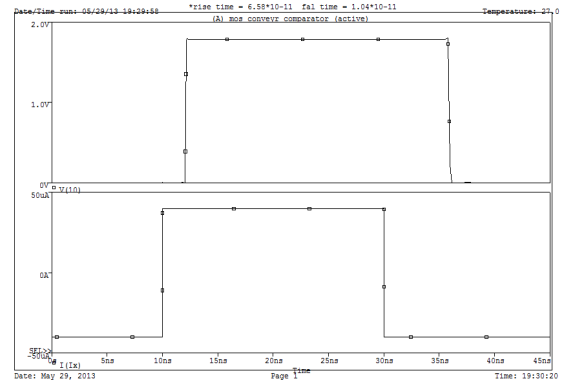


Fig.12 Simulation results for the proposed current comparator

Fig 12 shows the behaviour of the proposed circuit when a current pulse input injected in terminal X is in the range - 40 μA to 40 μA . The circuit exhibits a propagation delay of 4ns and power dissipation of 0.886 μW with current pulse in the range -40 μA to 40 μA .

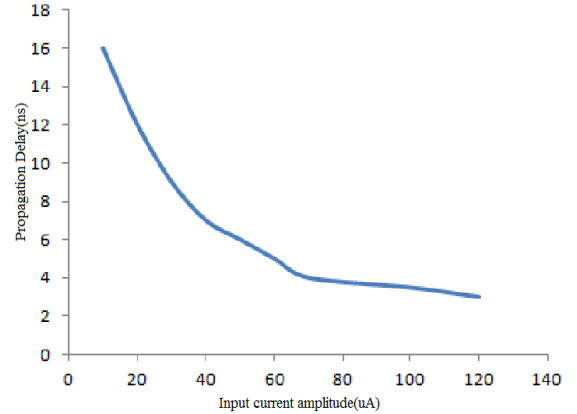


Fig.13 Variation in propagation delay with input current amplitude

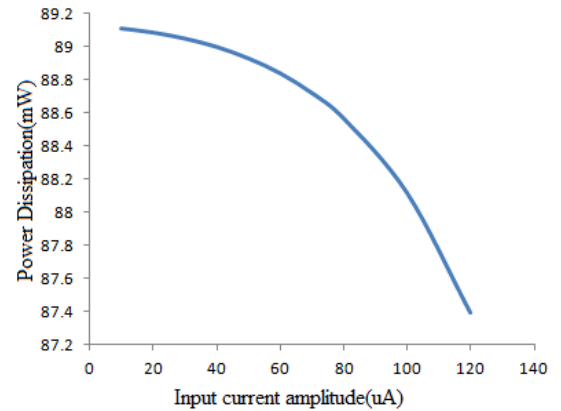


Fig. 14 Variation of power dissipation with input current amplitude.

Fig 13 and 14 show the variation of propagation delay and power dissipation with respect to input current respectively. It can be seen that propagation delay as well as power dissipation decrease with increase in input current amplitude. This decrease in propagation delay and power consumption with increase in input current amplitude is expected as with larger values of input current amplitude faster switching occurs.

A 2 bit ADC is simulated and Fig. 15 shows the behaviour of the ADC when a current pulse input is in the range $0\mu\text{A}$ to $30\mu\text{A}$ and 2- bit digital output i.e. B1B0 is coming out to be 01. ADC specifications (power dissipation, INL, DNL, offset error, area etc.) have been calculated and shown in Table 2. Also ADC ideal and actual transfer characteristics are shown in Fig. 16

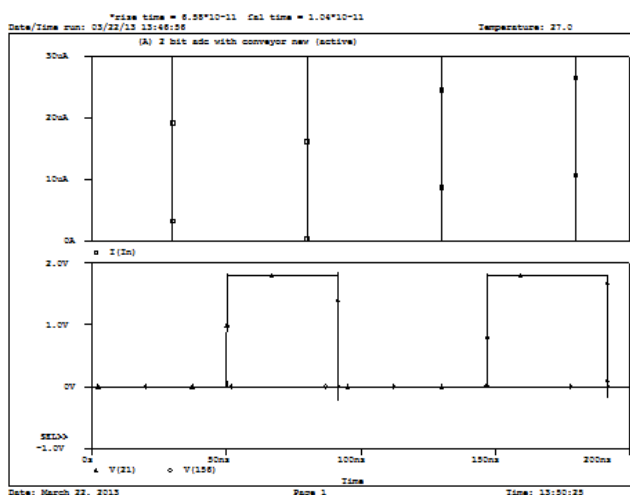


Fig.15 Simulation results for the 2-bit current mode flash ADC

TABLE II. SPECIFICATIONS OF 2-BIT CURRENT MODE FLASH ADC

Specifications	Values
INL	0.55 LSB
DNL	0.25 LSB
Propagation Delay I_{in} (0 to $30\mu\text{A}$) (0 to $51\mu\text{A}$) (0 to $80\mu\text{A}$)	16ns 11.5ns 10.5ns
Power Dissipation	286.33mW
Offset error	5uA
No of transistors	71
Resolution	4.4uA

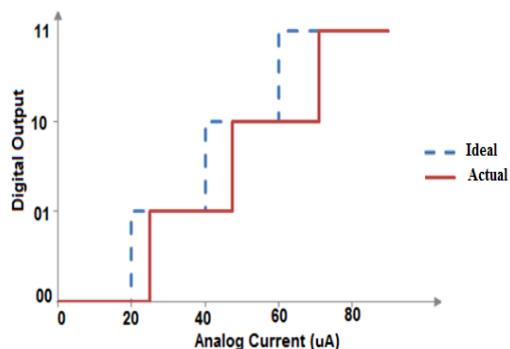


Fig 16 Ideal and actual ADC transfer characteristics

VI. CONCLUSION

A novel CMOS current comparator based on current conveyor II is proposed in the circuit. A positive feedback and op-amp is used in order to have relatively good performance and power dissipation, propagation delay and resolution has been calculated for the comparator. A 2 bit current mode flash ADC is designed and is found to perform satisfactorily.

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